

Business Intelligence Strategy for Improving Customer Satisfaction and Brand Perception at Breeze Airways

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This project centers on Breeze Airways, a United States-based carrier headquartered in Cottonwood Heights, Utah. Founded in 2021, the company has quickly distinguished itself as the fastest-growing carrier in America with over 60 aircraft in its fleet and counting. Its geographic market area is extensive for its age, serving over 85 cities across the United States, Mexico, and the Caribbean. Despite this rapid expansion, the airline faces significant infrastructure challenges that come with scaling operations so quickly. Currently, Breeze Airways struggles with a low customer recommendation rate. By using data and analytics from larger airlines to compare with Breeze, we aim to improve the company's performance.

For our data sources, we plan on using an Airline Reviews dataset and an Airline Passenger Satisfaction dataset, both from Kaggle. The Airline Reviews dataset provides fields that an airline would likely care about during rapid growth. Some fields include overall rating, review verification, service-category ratings such as seat comfort, cabin staff, ground service, Wi-Fi, value for money, etc, and even some other potentially useful fields. The dataset also includes the review itself in text format, which can help identify patterns in top complaints or even compliments. The Airline Passenger Satisfaction dataset provides fields all tied to customer satisfaction, including gate location, food and drink, online boarding, seat comfort, in-flight entertainment, baggage handling, and many more fields entirely revolving around how they affect customer satisfaction. Using this data, we built dashboards that analyze customer satisfaction trends over time and compare Breeze's ratings to those of the larger airlines within the same dataset. From this analysis, we can provide data-driven recommendations for how Breeze should focus their resources in order to support their rapid growth while maintaining/improving customer satisfaction.

Business Intelligence or BI, refers to the process of collecting, processing, analyzing, and interpreting data to help support business decisions. The BI process helps turn raw data into actionable data-driven insights by helping companies monitor company performance and identify key trends or patterns. For Breeze Airways, BI can generate value by helping the company better understand what factors influence passenger satisfaction and brand perception. Using BI, Breeze can get a better understanding of how their services are perceived by their customers and which areas they specifically need to improve upon.

The business problems that Breeze can address using BI technologies include understanding the customer experience and what influences their satisfaction the most, how customers typically perceive their brand, and having data spread out across multiple sources making it difficult to perform proper data analysis. Through BI, Breeze can work through centralizing the data and track important key performance indicators to determine what factors influence passenger satisfaction and brand perception.

The proposed BI framework that Breeze should use is similar to the one shown in class. The framework begins by first starting with the collection of both internal and external data sources. These internal sources include passenger satisfaction surveys, booking and

reservation systems, and customer loyalty program data. For external sources, airline review platforms and other publicly available feedback can be used. Once all this data is collected, it can then be processed using an ETL process to transform the data into a consistent format for easier analysis. This ensures that multiple systems can be combined and analyzed together in an effective manner. The processed data then needs to be stored within a centralized system including a data warehouse and data marts specialized to Breeze. The data warehouse would serve as the main storage unit for company data while the data marts would be specialized for specific analytical areas such as passenger satisfaction and review sentiments. Lastly, after storing the data, it can then be analyzed, interpreted, and visualized. The data is first analyzed to identify key trends or patterns, understand what topics are present in reviews, and monitor important key performance indicators. These insights can then be visualized using important metrics, charts, and filters to create a finalized interactive dashboard to help further examine performance across multiple segments.

The American division of Polish Airline LOT felt like they were flying blind when in 2016, their headquarters canceled their old BI subscription. With a fleet of around 100 aircraft and 9 billion in revenue in 2024. The Airline is swamped with data about day to flights customer spending, aircraft maintenance and customer reviews. The team eventually purchased AirlineBI, a Business intelligence software system from ARC, the BI company.

“Since the purchase of Airline BI less than two years ago, the tool has become an integral part of the airline’s U.S. sales team function and a guiding force in offering insight into its sales strategy. Using Airline BI, LOT Polish Airlines has improved performance measurement and agency reviews. The airline has also been able to improve its competitiveness on several routes with agency partners, as well as improve internal benchmarking on route performance, validating the addition of one nonstop route. Plus, the tool has saved the airline’s U.S. sales team approximately 15 percent more time on data compilation and analysis efforts alone. The airline’s Head of North American Sales Dominic Bernardo, now an Airline BI power user, claims the tool has provided his team with increased visibility into sales, particularly at the O&D level, and a quick and easy way to monitor all of their agency account activity – in turn, guiding insight into the airline’s indirect, U.S. sales strategy” (ARC Corp., n.d.).

“Since this initiative, the airline has increased its performance with this agency to seven routes, based on the positive results they received – an increase of 491 percent over the last 12 months, representing more than \$400,000 in growth” (ARC Corp., n.d.).

For a successful implementation of the proposed BI solution, Breeze should focus on first collecting reliable data either from their own internal sources or from platforms that contain some sort of verification for text reviews. Ensuring that the data is accurate is extremely important and reliable data generates reliable insights that can confidently be used for future business decisions. Additionally, to ensure that the implementation is successful, Breeze should provide managers, operational teams, and other higher-up executives easy access to the finalized dashboards and other analytical tools. This way, multiple levels at the company can access the data in real-time so the company can react quickly to not only any issues they notice, but common trends that may be heavily impacting their services.

In the future, this system could be extended through the use of machine learning and predictive analytics. The use of machine learning models could help in predicting which factors of a customer’s experience are most likely going to impact their overall satisfaction level or lead

to an overall negative experience with the company. Implementing these predictive analytics instead of just focusing on what has already happened, Breeze could be more proactive in improving their customer experience. This is especially important as the airline is small and actively growing at a rapid rate, so the use of machine learning models may give them an edge in their competition with all the other North American airlines.

For the Breeze to have a BI solution that is effective, the KPIs that are selected should be actionable and measurable. These metrics should be clearly defined and trackable to make sure that they directly support Breeze's goals, such as increasing its operational efficiency or improving overall customer loyalty. Also, having KPIs with high-level objectives such as raising overall ratings makes sure that every data point that is analyzed serves a broader purpose of supporting the airline's rapid growth.

Data quality is the foundation of any reliable BI system. Therefore, addressing issues such as null fields or inconsistent naming is a top priority. As a data management expert, I would implement standardized naming conventions and ensure all rating types are valid. To maintain data accuracy and completeness, I would use listwise deletion for missing rows to create a dataset that is universally applicable to any analytic techniques.

Breeze Airways is currently a North American airline meaning they are required to comply with the California Consumer Privacy Act (CCPA) when serving Californian passengers. While it is important to note that certain sections of the CCPA are not fully complete and may be changed by Californian legislation in the future the articles that are complete require: the right to opt out of sale of your data and sharing your data with third parties. The right to know what has been gathered from you. The right to delete your data and the right to correct possibly incorrect data.

Breeze can make use of social media analytics to help refine their business processes and strategic decisions. By monitoring platforms such as TripAdvisor and Kayak, the company can contextualize its own data with public sentiment. For example, sentiment analysis of customer reviews can be performed to identify aspects associated with positive and negative experiences customers have had with the airline. This can indicate what changes might need to be made to improve the overall customer experience and brand perception. Furthermore, tracking mentions on social media allows for improved customer engagement, enabling Breeze to address common complaints before they have the ability to escalate into broader issues.

Business Intelligence has shown to play an important role for some in the airline industry as companies continue to adopt this framework as a way to improve their data-driven insights. Breeze Airways is one of the fastest growing American Airlines and with its rapid growth, comes a large influx of data just waiting to be used. When professionally and correctly understood, this data can lead sustained company growth and growing customer experience. Our strategy details the creation of a data analysis pipeline which prioritizes customer behavior, customer's responses and operational data about our flights. Our dashboards indicate which metrics are key in influencing customer satisfaction and perception, helping Breeze understand in which areas they need to grow as a company to sustain themselves. With strong management oversight, data governance, and legal compliance with the proper regulations, our proposed BI analytical system will help Breeze strengthen their brand's reputation, improve overall customer experience, and sustain their continued growth in the airline industry.

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